Linear Algebra

norm: $\|x\|_p = (|x_1|^p + \dots + |x_n|^p)^{\frac{1}{p}}$

x is normalized if $\|x\|=1$

x, y are orthogonal $\Leftrightarrow x^T y = 0$

$U \in \mathbb{R}^{n\times n}$ is orthogonal if $U_i \perp U_j \ \forall i \neq j$

$U \in \mathbb{R}^{n\times n}$ is orthonormal if $\|U_i\|=1 \ \forall i$ and orthogonal

$\Rightarrow U^T U = I$

A is positive definite if $x^T A x > 0 \ \forall x \neq 0$

A is positive semi-definite if $x^T A x \geq 0 \ \forall x$

$Ax = \lambda x$

eigenvector, eigenvalue

zero is an eigenvalue $\Leftrightarrow$ singular $\Leftrightarrow$ invertible $\Leftrightarrow \det(A) = 0$

$\boxed{\text{SVD}}$ $A \in \mathbb{R}^{n\times m}$ $U \in \mathbb{R}^{n\times n}$ $V \in \mathbb{R}^{m\times m}$

$A = U \Sigma V^T$ ($\Rightarrow AV = U\Sigma$)

* U, V are orthonormal
* singular vals of A
  - diag elements of Σ
  - $= \text{sqrt}(\text{eigenvals of } A^TA, AA^T)$
* cols of U are eigen vectors of AA^T
* cols of V are eigen vectors of A^TA

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{pmatrix} \quad f: \mathbb{R}^n \to \mathbb{R}$$

$$\nabla^2 f(x) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f}{\partial x_1 x_n} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}$$

Convexity

f is convex if $\forall x, y$

$f\left(\frac{x+y}{2}\right) \leq \frac{f(x)}{2} + \frac{f(y)}{2}$

* If f is differentiable

$f(x+y) \geq f(x) + \nabla_x f(x) \cdot y$

* If f is twice $\sim$

$\nabla^2 f(x)$ is positive semi-definite

$\nabla_x (b^T x) = \nabla_x (x^T b) = b$

$\Rightarrow \nabla_x (y^T A x) = y^T A$

$\nabla_y (y^T A x) = Ax$

$\nabla_x (Ax) = A \quad \nabla_x (x^T A) = A^T$

$\nabla_x (x^T A x) = 2Ax$

$\nabla_x^2 (x^T A x) = 2A$

$\nabla_x (\|x\|^2) = \nabla_x (x^T x) = 2x$

$\nabla_x (x) = \nabla_x (x^T) = I$

Let a, b, z be function of x

$\nabla_x (Az) = A \nabla_x (z)$

$\nabla_x (a^T b) = a^T \nabla_x (b) + b^T \nabla_x (a)$

$x^T A x = \sum_i \sum_j A_{ij} x_i x_j$

Lagrange

problem: minimize $f(x)$

subject to $g_i(x) \leq 0 \ \forall i$

$h_j(x) = 0 \ \forall j$

$L(x, \lambda, \nu) = f(x) + \sum_i \lambda_i g_i(x) + \sum_j \nu_j h_j(x)$

$L_D(\lambda, \nu) = \min_x L(x, \lambda, \nu)$

maximize $L_D(\lambda, \nu)$

subject to $\lambda_i, \nu_j \geq 0$

Probability

Conditional $\sim$

$P(X|Y) = P(X, Y)/P(Y)$

independent

$P(X, Y) = P(X)P(Y)$

$P(X|Y) = P(X)$

Conditionally independent

$P(X, Y|Z) = P(X|Z)P(Y|Z)$

$P(X|Y, Z) = P(X|Z)$

marginal $\sim$

$P(X) = \sum_y P(X|Y=y)P(y)$

$Var[X] = E[X - E[X]]^2$

$= E[X^2] - E[X]^2$

$Var[af(x)] = a^2 Var[f(x)]$

$Var[X+Y] = Var[X] + Var[Y] + 2Cov[X, Y]$

$Cov[X, Y] = E[(X - E[X])(Y - E[Y])]$

$= E[XY] - E[X]E[Y]$

Bayes' Rule

$$P(X|Y) = \frac{P(Y|X)P(X)}{P(Y)}$$

$$P(X_i|Y) = \frac{P(Y|X_i)P(X_i)}{\sum P(Y|X_i)P(X_i)}$$